## Taxonomic screening and reference validation of non-target reads in Populus trichocarpa

### Objective and methods

Potential non-target sequences were evaluated in NCBI SRA datasets SRR36001243 (DNA WGS; 100,000 paired 151-bp fragments) and SRR3241527 (RNA-seq; 100,000 single-end 100-bp reads). Reads were screened with Kraken2 (PlusPFP-8, 26 February 2026). Selected taxonomic assignments were extracted and validated by Bowtie2 mapping to Populus trichocarpa (GCF_000002775.5), human GRCh38.p14 (GCF_000001405.40), and selected microbial assemblies. SAMtools was used to assess mapping quality, alignment locations, and reference breadth/depth. Three RNA human-candidate reads were also tested against 59,542 annotated Populus transcripts.

### Results

| Library / candidate | Kraken2 assignments | Independent validation / provisional bin |
|---|---|---|
| DNA · Human | 124 pairs (0.124%) | 0 human, 200 Populus-aligned individual reads; noise |
| DNA · Wolbachia | 4 pairs (0.004%) | 8/8 reads favor Populus; four pairs share one fragment location; noise |
| RNA · Human | 15 reads (0.015%) | 3 human-preferred reads (0.003% of screened reads); possible trace |
| RNA · Iamia | 2 reads (0.002%) | 2/2 favor Populus; likely noise |
| RNA · Kosakonia | 1 read (0.001%) | 1/1 favors Populus; likely noise |

RNA human-specific coverage (selected candidate subset): 291 aligned bases over a 3,298,430,636-bp human reference; breadth = 0.0000088224%; mean depth = $8.8224 \times 10^{-8}\times$ (MAPQ $\geq$ 20 and base quality $\geq$ 20). All three reads had NM = 0 (two 100M alignments; one 91M9S); none mapped to annotated Populus transcripts under the tested settings. DNA human-reference breadth and depth were zero for the selected candidate subset.

### Interpretation

The DNA human calls and the three tested microbial taxa were predominantly supported by host-genome alignments, indicating likely taxonomic classification artifacts rather than confirmed contaminants. Wolbachia-classified fragments also shared identical paired-end coordinates, suggesting duplicate-like observations. In contrast, three RNA reads aligned to distinct human genomic regions with MAPQ 40–44 and no aligned-base mismatches, constituting a possible trace human-origin signal; its source and biological significance remain unconfirmed. No consequential contamination was established.

### Limitations and conclusion

The analysis used the first 100,000 spots/reads—not random or whole-run samples—and targeted validation of selected Kraken2 hits. Coverage metrics characterize selected reads rather than full-library contaminant abundance. Unrepresented strains, incomplete host transcript annotation, ambiguous alignments, and missing negative controls constrain inference. Overall, independent reference validation substantially reduced apparent microbial detections while leaving a small, provisional human-aligned RNA signal requiring broader and orthogonal verification.